

Problem 15.14

A family wishes to accumulate \$50,000 in a college education fund by the end of 20 years. If they deposit \$1,000 into the fund at the end of each of the first 10 years, and \$1,000 + X at the end of each of the second 10 years, find X to the nearest unit if the fund earns 7% effective.

Solution

The annual effective interest rate is

$$i = 0.07.$$

We accumulate all deposits to time 20.

Deposits from years 1–10

The first ten deposits form an annuity-immediate of \$1,000 accumulated to time 10, then accumulated for another 10 years to time 20. Its value at time 20 is

$$1000s_{\overline{10}|}(1.07)^{10}.$$

Deposits from years 11–20

The second ten deposits form an annuity-immediate of \$1,000 + X accumulated to time 20. Its value at time 20 is

$$(1000 + X)s_{\overline{10}|}.$$

Therefore, the accumulated value equation is

$$1000s_{\overline{10}|}(1.07)^{10} + (1000 + X)s_{\overline{10}|} = 50000.$$

Solving for X gives

$$X = \frac{50000 - 1000s_{\overline{10}|}(1.07)^{10}}{s_{\overline{10}|}} - 1000.$$

Using

$$s_{\overline{n}|} = \frac{(1+i)^n - 1}{i},$$

we get

$$s_{\overline{10}|} = \frac{(1.07)^{10} - 1}{0.07} \approx 13.8164.$$

Thus,

$$\begin{aligned} X &= \frac{50000 - 1000(13.8164)(1.07)^{10}}{13.8164} - 1000 \\ &\approx 651.7238. \end{aligned}$$

Hence, to the nearest unit,

$$\boxed{X \approx 652}.$$